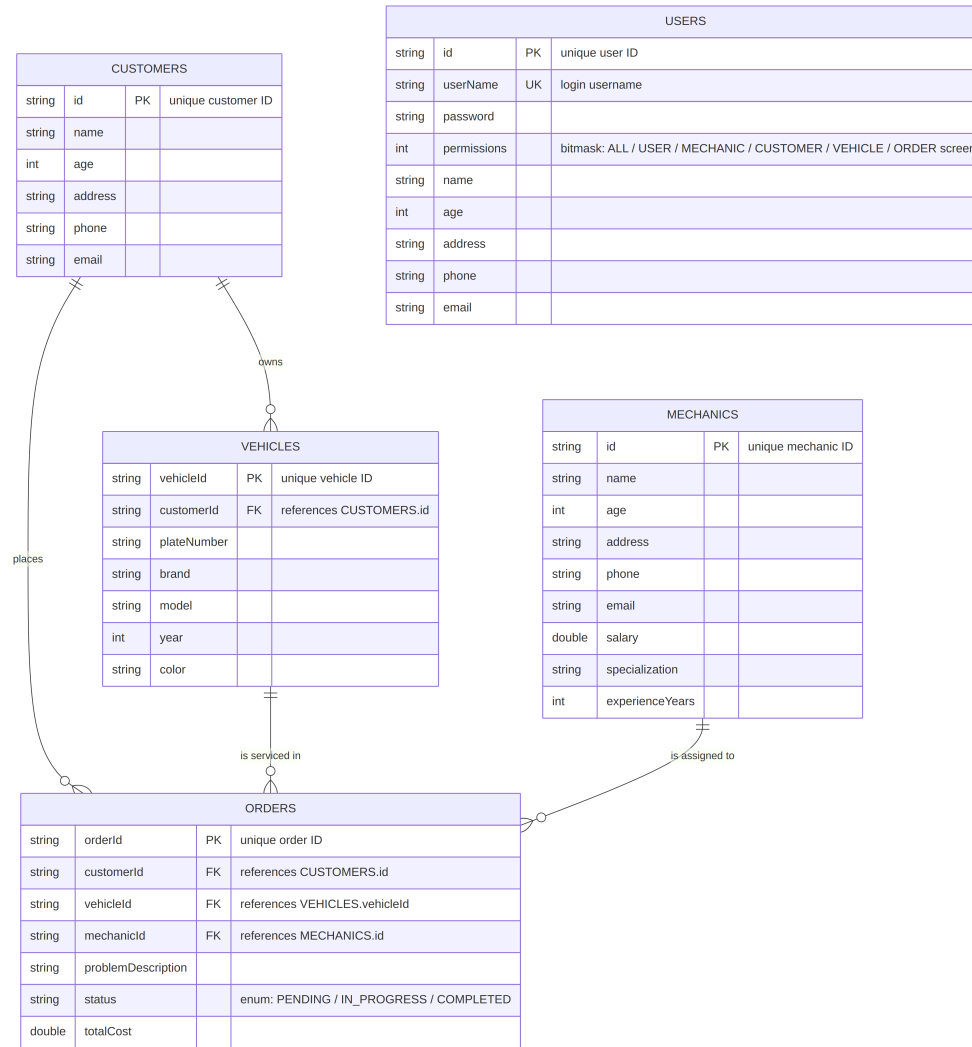


Maintenance Management System

Entity-Relationship Diagram (ERD)



Entities and attributes

Entity	Attribute	Type	Key	Notes
CUSTOMERS	id	string	PK	Unique customer ID
	name	string		
	age	int		
	address	string		
	phone	string		
	email	string		
MECHANICS	id	string	PK	Unique mechanic ID
	name	string		
	age	int		
	address	string		
	phone	string		
	email	string		
	salary	double		
	specialization	string		
	experienceYears	int		
VEHICLES	vehicleId	string	PK	Unique vehicle ID
	customerId	string	FK	References CUSTOMERS.id
	plateNumber	string		
	brand	string		
	model	string		
	year	int		
	color	string		

Entity	Attribute	Type	Key	Notes
ORDERS	orderId	string	PK	Unique order ID
	customerId	string	FK	References CUSTOMERS.id
	vehicleId	string	FK	References VEHICLES.vehicleId
	mechanicId	string	FK	References MECHANICS.id
	problemDescription	string		
	status	string (enum)		PENDING / IN_PROGRESS / COMPLETED
	totalCost	double		
USERS	id	string	PK	Unique user ID
	userName	string	UK	Login username (must be unique)
	password	string		
	permissions	int (bitmask)		ALL(-1) / USER(1) / MECHANIC(2) / CUSTOMER(4) / VEHICLE(8) / ORDER(16)
	name	string		
	age	int		
	address	string		
	phone	string		
	email	string		

Relationships

Relationship	Cardinality	Description
CUSTOMERS -> VEHICLES	1 to many	A customer can own multiple vehicles; each vehicle belongs to exactly one customer.
CUSTOMERS -> ORDERS	1 to many	A customer can place multiple maintenance orders.
VEHICLES -> ORDERS	1 to many	A vehicle can be linked to multiple orders over time (repeat visits).

MECHANICS -> ORDERS	1 to many	A mechanic can be assigned to multiple orders; each order has exactly one mechanic.
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Implementation notes

- USERS is a standalone entity used for system login and screen-level permissions (via a bitmask on the **permissions** field). No foreign key currently links it to CUSTOMERS or MECHANICS in the codebase.
- All entities inherit shared personal fields (name, age, address, phone, email) from a common abstract **Person** class; MECHANICS additionally inherits salary and id from an abstract **Employee** class. These are code-level inheritance, not separate database tables.
- Data is persisted as flat delimited text files (one per entity: Customers.txt, Mechanics.txt, Vehicles.txt, Orders.txt, Users.txt) rather than a relational database engine; the keys and relationships above reflect the logical schema implied by the code.